

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular & Supplementary Winter Examination – 2023 Course: B. Tech. Branch: All B.tech Branches Semester : I Subject Code & Name: BTES105 Energy and Environment Engineering Max Marks: 60 Date:10-01-24 Duration: 3 Hr.		
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain in details Nuclear power plant and write the functions of a) moderator b) fuel rod and c) control rod.	Remember/ CO1	6
B)	Explain working of open cycle gas turbine power plant with neat sketch. Write advantages and limitations of gas turbine power plant.	Remember/CO1	6
C)	What is thermoelectric generator? Explain construction and working of thermionic power generator.	Remember/CO1	6
Q.2	Solve Any Two of the following.		12
A)	Compare the relative characteristic of HAWT and VAWT. List the major application of wind power.	Understand/CO2	6
B)	Define solar energy with suitable example. Explain working of parabolic collector. Write advantages of photovoltaic system.	Remember/CO2	6
C)	What is the Biomass? Explain construction and working of biogas plant with neat sketch.	Remember/CO2	6
Q. 3	Solve Any Two of the following.		12
A)	What do you mean by energy efficiency? Explain the measures to be taken to reduce the energy conservation in compressor.	Remember/CO3	6
B)	Discuss the scope of energy conservation and principle of energy conservation. How you conserve energy in heat producing equipment with suitable example.	Remember/CO3	6
C)	How will you increase energy efficiency in boiler? Also explain the energy conservation in lightning system.	Remember/CO3	6
Q.4	Solve Any Two of the following.		12

A)	Why TajMahal is turning yellow? Write importance and objectives of National Ambient Air Quality Standards.	Understand/CO4	6
B)	What is a Smog? Explain photochemical smog. Write effect of photochemical smog on human, plant & materials.	Remember/CO4	6
C)	List the control methods of particulate matter. Explain any one with suitable sketch.	Remember/CO4	6
Q. 5	Solve Any Two of the following.		12
A)	Define Noise pollution. What are the effect of noise pollution on human, wildlife and properties?	Remember/CO5	6
B)	What are the main causes of river pollution? How can river pollution to be controlled?	Remember/CO5	6
C)	How soil pollution occurs? Write control measures of soil pollution.	Remember/CO5	6
	*** End ***		

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